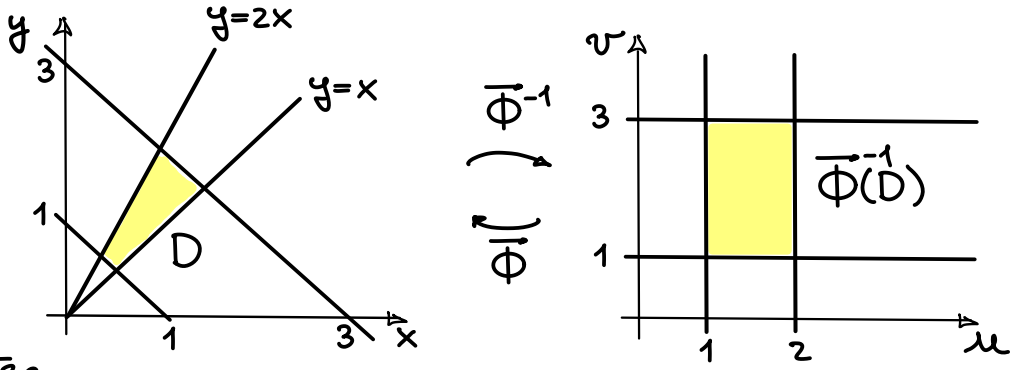
ANALISI MATEMATICA 2 - LEZIONE 19

ESEMPI

- $\iint_D \frac{1}{xy}\,dx\,dy \quad D=\{(x,y): 0\le x\le y\le 2x,\ 1\le x+y\le 3\}.$

Poniamo

$$\begin{cases} u=\frac{y}{x} \\ v=x+y \end{cases} \quad \Phi^{-1}(x,y) \quad \text{da cui} \quad \begin{cases} x=\frac{v}{u+1} \\ y=\frac{uv}{u+1} \end{cases} \quad \Phi(u,v)$$



Inoltre

$$\det(J_\Phi(u,v))=\det\left(\begin{bmatrix} -\frac{v}{(u+1)^2} & \frac{1}{u+1} \\ \frac{v}{(u+1)^2} & \frac{u}{u+1} \end{bmatrix}\right)=\frac{-vu-v}{(u+1)^3}=\frac{-v}{(u+1)^2}$$

Allora

$$\iint_D \frac{1}{xy}\,dx\,dy=\iint_{\Phi^{-1}(D)} \frac{u+1}{v}\cdot\frac{u+1}{uv}\cdot\left|\frac{-v}{(u+1)^2}\right|du\,dv$$

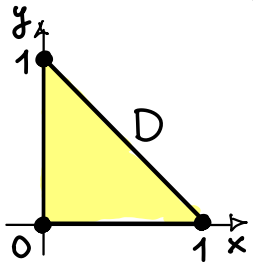
$$=\int_{u=1}^{2}\left(\int_{v=1}^{3}\frac{(u+1)^2}{uv^2}\cdot\frac{v}{(u+1)^2}\,dv\right)du$$

$$=\int_1^2 \frac{1}{u}\left[\log(v)\right]_1^3 du$$

$$=\log(3)\cdot\left[\log(u)\right]_1^2=\log(3)\cdot\log(2).$$

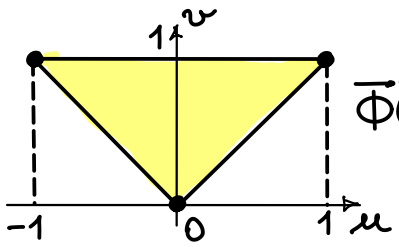
- $\iint_D \frac{(x-y)^2}{1+x+y}\,dx\,dy \qquad D=\{(x,y): x\geq 0, y\geq 0, x+y\leq 1\}.$

Poniamo $\begin{cases} u=x-y \\ v=x+y \end{cases}$ $\vec{\Phi}^{-1}(x,y)$ da cui $\begin{cases} x=\frac{u+v}{2} \\ y=\frac{v-u}{2} \end{cases}$ $\vec{\Phi}(u,v)$



$\vec{\Phi}^{-1}$ ⟶

⟵ $\vec{\Phi}$



$\vec{\Phi}$
$(0,0) \to (0,0)$
$(1,1) \to (1,0)$
$(-1,1) \to (0,1)$
$\vec{\Phi}$ è lineare

Inoltre

$$\det(J_{\vec{\Phi}}(u,v)) = \det\left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\right) = \frac{1}{4}+\frac{1}{4} = \frac{1}{2}.$$

Allora

$$\iint_D \frac{(x-y)^2}{1+x+y}\,dx\,dy = \iint_{\vec{\Phi}^{-1}(D)} \frac{u^2}{1+v}\left|\frac{1}{2}\right| du\,dv$$

u-semplice ↘

$$= \frac{1}{2}\int_{v=0}^{1}\left(\int_{u=-v}^{v}\frac{u^2}{1+v}\,du\right)dv$$

$$= \frac{1}{2}\int_{v=0}^{1}\frac{1}{1+v}\left[\frac{u^3}{3}\right]_{-v}^{v}dv$$

$$= \frac{1}{3}\int_0^1 \frac{v^3}{1+v}\,dv$$

$$= \frac{1}{3}\int_0^1\left(1-v+v^2-\frac{1}{1+v}\right)dv$$

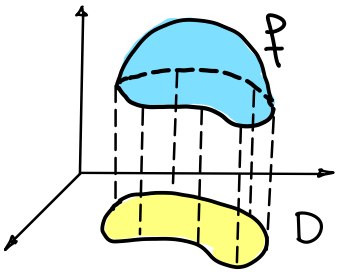
$$= \frac{1}{3}\left[v-\frac{v^2}{2}+\frac{v^3}{3}-\log(1+v)\right]_0^1$$

$$= \frac{1}{3}\left(1-\frac{1}{2}+\frac{1}{3}-\log(2)\right) = \frac{5}{18}-\frac{\log(2)}{3}.$$

OSSERVAZIONE

Se $f(x,y) \geq 0$ per ogni $(x,y) \in D$ allora

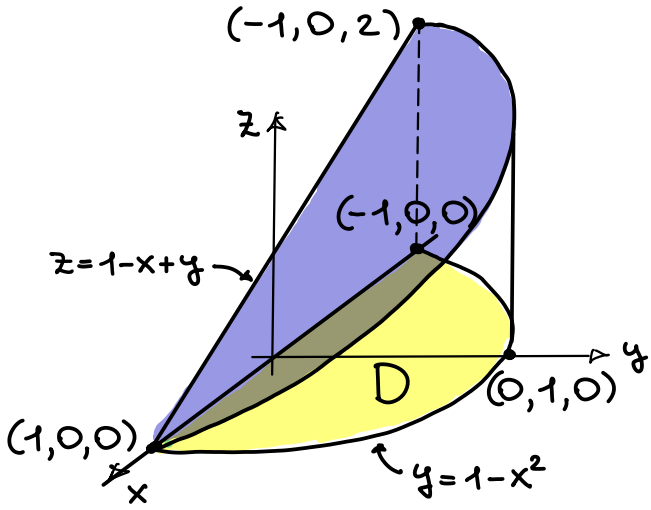
$$\iint_D f(x,y)\,dx\,dy = \text{Volume}(E) = |E|$$



dove $E = \{(x,y,z) : (x,y) \in D,\ 0 \leq z \leq f(x,y)\}$.

- Calcolare il volume di

$$E = \{(x,y,z) : y \geq 0,\ z \geq 0,\ x - y + z \leq 1,\ y \leq 1 - x^2\}$$



Consideriamo

$$D = \{(x,y) : 0 \leq y \leq 1 - x^2\}$$

e

$$z = f(x,y) = 1 - x + y.$$

Si noti che se $(x,y) \in D$ allora

$$f(x,y) \geq 1 - x \geq 0.$$

($f(x,y) \geq 1 - x$ perché $y \geq 0$; $1 - x \geq 0$ perché $x \leq 1$)

$$|E| = \iint_D (1 - x + y)\,dx\,dy = \int_{x=-1}^{1} \left(\int_{y=0}^{1-x^2} (1 - x + y)\,dy \right) dx$$

($[-1,1]$ è simmetrico)

$$= \int_{-1}^{1} \left[y - xy + \frac{y^2}{2} \right]_0^{1-x^2} dx = \int_{-1}^{1} \left((1 - x^2) - x(1 - x^2) + \frac{(1 - x^2)^2}{2} \right) dx$$

(pari, dispari, pari)

$$= \int_0^1 \left(2(1 - x^2) + (1 - 2x^2 + x^4) \right) dx = \int_0^1 (3 - 4x^2 + x^4)\,dx$$

$$= \left[3x - \frac{4x^3}{3} + \frac{x^5}{5} \right]_0^1 = \frac{45 - 20 + 3}{15} = \frac{28}{15}.$$

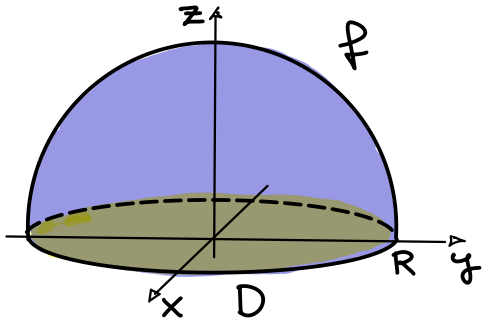
- Calcolare il volume di una sfera di raggio R.

La sfera di raggio R e centro $(0,0,0)$ ha equazione

$$x^2+y^2+z^2=R^2.$$

Sia $E=\{(x,y,z) : x^2+y^2+z^2\leq R^2\}$ e consideriamo la semisfera data dal grafico della funzione

$$z=f(x,y)=\sqrt{R^2-x^2-y^2} \quad \text{in} \quad D=\{(x,y) : x^2+y^2\leq R^2\}.$$



Allora

$$|E|=2\iint_D f(x,y)\,dx\,dy$$

$$=2\iint_{\{x^2+y^2\leq R^2\}} \sqrt{R^2-x^2-y^2}\,dx\,dy$$

$$\overset{CP}{=} 2\iint_{[0,R]\times[0,2\pi]} \sqrt{R^2-\rho^2}\,\rho\,d\rho\,d\theta$$

$$=2\int_{\rho=0}^{R}\sqrt{R^2-\rho^2}\left(\int_{\theta=0}^{2\pi} d\theta\right)\rho\,d\rho$$

$$=4\pi\int_0^R \sqrt{R^2-\rho^2}\,\rho\,d\rho$$

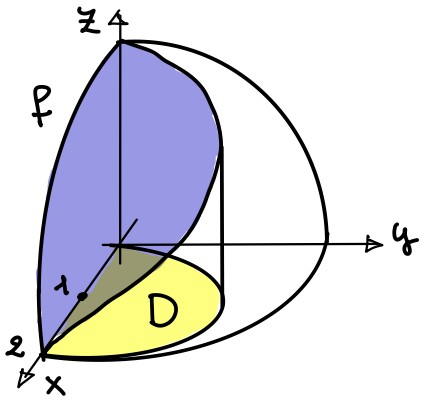
$t=R^2-\rho^2$
$dt=-2\rho\,d\rho$

$$\overset{2}{=} 4\pi\int_{R^2}^{0}\sqrt{t}\left(\frac{dt}{-2}\right)=2\pi\left[\frac{2t^{3/2}}{3}\right]_0^{R^2}=\frac{4}{3}\pi R^3.$$

- Calcolare il volume dell'insieme

$$E=\{(x,y,z) : x,y,z\geq 0,\ x^2+y^2+z^2\leq 4,\ x^2+y^2\leq 2x\}$$

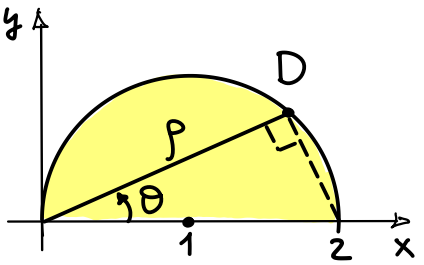
sfera $\quad$ $\hookrightarrow (x-1)^2+y^2\leq 1$ cilindro



Consideriamo la funzione

$$z=f(x,y)=\sqrt{4-x^2-y^2}$$

in $D=\{(x,y): x^2+y^2\leq 2x,\ y\geq 0\}$.



In coordinate polari

$$\overline{\Phi}^{-1}(D)=\left\{(\rho,\theta): \begin{array}{l} 0\leq\rho\leq 2\cos\theta \\ \theta\in[0,\frac{\pi}{2}] \end{array}\right\}$$

Allora

$$|E|=\iint_D \sqrt{4-x^2-y^2}\,dx\,dy \overset{CP}{=} \iint_{\overline{\Phi}^{-1}(D)} \sqrt{4-\rho^2}\cdot\rho\,d\rho\,d\theta$$

$$=\int_{\theta=0}^{\frac{\pi}{2}}\left(\int_{\rho=0}^{2\cos\theta}\sqrt{4-\rho^2}\cdot\rho\,d\rho\right)d\theta$$

$t=4-\rho^2$, $dt=-2\rho\,d\rho$

$$=\int_0^{\frac{\pi}{2}}\left(\int_{t=4}^{4\sin^2\theta}\sqrt{t}\left(-\frac{dt}{2}\right)\right)d\theta=\int_0^{\frac{\pi}{2}}\left[\frac{t^{3/2}}{3}\right]_{4\sin^2\theta}^{4}d\theta$$

$$=\frac{8}{3}\int_0^{\frac{\pi}{2}}(1-\sin^3\theta)\,d\theta$$

$$=\frac{8}{3}\cdot\frac{\pi}{2}+\frac{8}{3}\int_0^{\frac{\pi}{2}}(1-\cos^2\theta)\,d(\cos\theta)$$

$$=\frac{4\pi}{3}+\frac{8}{3}\left[\cos\theta-\frac{\cos^3\theta}{3}\right]_0^{\frac{\pi}{2}}=\frac{4\pi}{3}-\frac{16}{9}.$$

- Calcolare

1) $\lim_{n\to\infty} \iint_{D_n} e^{-(x^2+y^2)}\,dx\,dy$ con $D_n=\{(x,y): x^2+y^2\le n^2\}$.

2) $\lim_{n\to\infty} \iint_{Q_n} e^{-(x^2+y^2)}\,dx\,dy$ con $Q_n=[-n,n]\times[-n,n]$.

1) $$\iint_{D_n} e^{-(x^2+y^2)}\,dx\,dy \overset{CP}{=} \int_{\rho=0}^{n} e^{-\rho^2}\left(\int_{\vartheta=0}^{2\pi} d\vartheta\right)\rho\,d\rho$$
$$= 2\pi\int_0^n e^{-\rho^2}\rho\,d\rho = \cancel{2}\pi\left[-\frac{e^{-\rho^2}}{\cancel{2}}\right]_0^n = \pi(1-e^{-n^2}) \xrightarrow{n\to\infty} \pi.$$

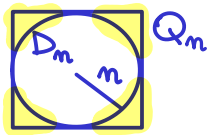
2) $$\iint_{Q_n} e^{-(x^2+y^2)}\,dx\,dy = \int_{x=-n}^{n} e^{-x^2}\cdot\left(\int_{y=-n}^{n} e^{-y^2}\,dy\right)dx$$ ← non dipende da x
$$= \left(\int_{-n}^{n} e^{-x^2}dx\right)\cdot\left(\int_{-n}^{n} e^{-y^2}dy\right) \xrightarrow{n\to\infty} \left(\int_{-\infty}^{+\infty} e^{-x^2}dx\right)^2.$$

OSSERVAZIONE

I due limiti ottenuti in 1) e 2) sono uguali? SÌ

$$0 \le \iint_{Q_n} e^{-(x^2+y^2)}\,dx\,dy - \iint_{D_n} e^{-(x^2+y^2)}\,dx\,dy = \iint_{Q_n\setminus D_n} e^{-(x^2+y^2)}\,dx\,dy$$

$Q_n \supseteq D_n$ e $f\ge 0$



$$\le \sup_{Q_n\setminus D_n} e^{-(x^2+y^2)} \cdot |Q_n\setminus D_n| = e^{-n^2}\cdot(4n^2-\pi n^2)$$

∂D_n

$\to 0$ per $n\to\infty$

Quindi $\pi = \left(\int_{-\infty}^{+\infty} e^{-x^2}dx\right)^2$ ossia $\int_{-\infty}^{+\infty} e^{-x^2}dx = \sqrt{\pi}$ che NON si può ottenere integrando direttamente perchè le primitive di e^{-x^2} non sono esprimibili mediante funzioni elementari.